

ZIJUN WANG

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🎓 EDUCATION

University of California, Santa Cruz, Santa Cruz, United States 2024.08 – Present

PhD student (Expected graduation date: 06/2028)

Advised by Prof. Cihang Xie at VLAA LAB in Computer Science and Engineering Department

Research interest: Autonomous Agents, Data-Centric LLM Training, AI Safety & Alignment

Zhejiang University, Hangzhou, China 2020.09 – 2024.06

Bachelor of Engineering

Major in Computer Science and Technology, College of Computer Science and Technology

🎓 EXPERIENCE

Amazon Intern Palo Alto, CA

Continue Learning ads 2026.06 – Present

- Under Supervision of Hanqing Lu
- Worked on **RSI Agent / Research World Model**
- Co-authored **Harness Updating Is Not Harness Benefit** (Accepted by EMNLP)

ByteDance Intern San Jose, CA

Data-TnS-Algo-Foundations & Intelligence Service 2025.06 – 2026.04

- Under Supervision of Fengze Liu
- Worked on **Pretraining Foundation LLM**
- First-authored **NAG-based Ranking**, a training-free and interpretable framework for target-oriented pre-training data selection (Accepted by **ICML 2026**)
- Co-authored **InfoLaw**, a data-aware scaling law framework modeling quality-weighted mixture data and repetition (Accepted by **ICML 2026**)

🎓 PUBLICATIONS

Your Agent, Their Asset: A Real-World Safety Analysis of OpenClaw

Zijun Wang, et al.

Accepted by *Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*

TL;DR: A real-world safety analysis of OpenClaw, a widely-used personal AI agent with broad system access (Gmail, Stripe, etc.). We introduce the **CIK taxonomy** (Capability, Identity, Knowledge) for persistent agent state, and show that poisoning any single CIK dimension raises attack success rate from 24.6% to 64–74% across four models; even the strongest defense still admits 63.8%, indicating the vulnerabilities are architectural.

Harness Updating Is Not Harness Benefit: Disentangling Evolution Capabilities in Self-Evolving LLM Agents

Minhua Lin, Juncheng Wu, Zijun Wang, et al.

Accepted by *Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*

TL;DR: We disentangle self-evolving LLM agents into **harness-updating** (producing persistent prompt, skill, memory, and tool updates from execution evidence) and **harness-benefit** (using those updates during task solving). Updating quality is surprisingly flat across base-model tiers, while benefit is non-monotonic: weak models struggle to invoke or follow updated harnesses, mid-tier models benefit most, and strong models gain less.

VisualClaw: A Real-Time, Personalized Agent for the Physical World

Haoqin Tu, Jianwen Chen, **Zijun Wang**, et al.

In submission to *Conference on Neural Information Processing Systems (NeurIPS 2026)*

TL;DR: VisualClaw is a self-evolving multimodal agent for real-time physical-world video tasks, combining hybrid encoding (streaming frame gating and hot/cold skill-bank injection) with failure-driven skill evolution. Across four video-QA benchmarks, it cuts per-question API cost by 98% versus full-frame upload while improving accuracy in most settings, and introduces VisualClawArena, a 200-scenario multimodal agentic benchmark.

VLAA-GUI: Knowing When to Stop, Recover, and Search, A Modular Framework for GUI Automation

Qijun Han, Haoqin Tu, **Zijun Wang**, et al.

In submission to *Conference on Neural Information Processing Systems (NeurIPS 2026)*

TL;DR: VLAA-GUI is a modular GUI agentic framework integrating a Completeness Verifier (visual-evidence success validation), a Loop Breaker (multi-tier filtering against repetitive failures), and a Search Agent (LM-queried workflows). It achieves 77.5% on OSWorld and 61.0% on WindowsAgentArena, with three backbones surpassing human performance on OSWorld in a single pass.

STAR-1: Safer Alignment of Reasoning LLMs with 1K Data

Zijun Wang, et al.

Accepted by *Association for the Advancement of Artificial Intelligence (AAAI 2026 (oral))*

TL;DR: STAR-1 is a high-quality, just-1k-scale safety dataset for LLMs built on diversity, deliberative reasoning, and rigorous filtering. Fine-tuning LLMs with STAR-1 leads to an average 40% improvement in safety across four benchmarks, with only a marginal 1.1% average decrease in reasoning ability across five tasks.

Target-Oriented Pretraining Data Selection via Neuron-Activated Graph

Zijun Wang, et al.

Accepted by *International Conference on Machine Learning (ICML 2026)*

TL;DR: We introduce Neuron-Activated Graph Ranking (NAG), a training-free and interpretable framework for target-oriented pretraining data selection. NAG characterizes each target input by a sparse set of high-impact neurons in off-the-shelf LLMs, improving target-oriented pretraining by 4.9% over random sampling and outperforming SOTA baselines by 5.3% on HellaSwag; deactivating just 0.12% of the identified neurons causes substantial degradation, isolating a sparse functional backbone for target-specific features.

AttnGCG: Enhancing Adversarial Attacks on Language Models with Attention Manipulation

Zijun Wang, et al.

Accepted by *Transactions on Machine Learning Research (TMLR 2025)*

TL;DR: AttnGCG additionally manipulates models' attention scores to enhance LLM jailbreaking, achieving an average improvement of 7% in Llama-2 and 10% in Gemma series, with robust attack transferability against unseen harmful goals and black-box LLMs.

How Many Unicorns Are in This Image? A Safety Evaluation Benchmark for Vision LLMs

Haoqin Tu*, Chenhang Cui*, **Zijun Wang***, et al. (* equal contribution)

Accepted by *European Conference on Computer Vision (ECCV 2024)*

TL;DR: A comprehensive safety evaluation suite for Vision LLMs, shifting focus from standard performance to out-of-distribution generalization and adversarial robustness.

MIRA: When Visualizing is the First Step to Reasoning, a Benchmark for Visual Chain-of-Thought

Yiyang Zhou*, Haoqin Tu*, **Zijun Wang**, et al. (* equal contribution)

Accepted by *Conference on Computer Vision and Pattern Recognition (CVPR 2026)*

TL;DR: MIRA is a benchmark where generating intermediate visual images (sketches, diagrams, path drawings) is essential for reasoning, targeting tasks with complex structures and spatial relationships that are difficult to express through language alone.

Chasing the Public Score: User Pressure and Evaluation Exploitation in Coding Agent Workflows

Accepted by *Conference on Language Modeling (COLM 2026)*

TL;DR: Examines whether coding agents exploit evaluation metrics under user pressure. Using AgentPressureBench (34 tasks, 1,326 trajectories from 13 agents), identifies 403 exploitative behaviors; stronger models exhibit higher exploitation rates (correlation 0.77), and explicit anti-exploit prompting reduces exploitation from 100% to 8.3%.

InfoLaw: Information Scaling Laws for Large Language Models with Quality-Weighted Mixture Data and Repetition

Accepted by *International Conference on Machine Learning (ICML 2026)*

TL;DR: A data-aware scaling framework that predicts LLM loss from model size, data mixture weights, and repetition, enabling efficient data-recipe selection under varying compute budgets.

AHELM: A Holistic Evaluation of Audio-Language Models

Tony Lee, Haoqin Tu, Chi Heem Wong, Zijun Wang, et al.

arXiv 2025

TL;DR: A comprehensive benchmark for evaluating audio-language models across diverse audio understanding tasks.

Where on Earth? A Vision-Language Benchmark for Probing Model Geolocation Skills Across Scales

arXiv 2025

TL;DR: A benchmark probing VLM geolocation capabilities across multiple spatial scales.

Mimicking the Physicist’s Eye: A VLM-centric Approach for Physics Formula Discovery

arXiv 2025

TL;DR: Leveraging vision-language models to automate physics formula discovery from visual observations.

Handling Feature Heterogeneity with Learnable Graph Patches

Yifei Sun, Yang Yang, Xiao Feng, Zijun Wang, et al.

Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2025)

TL;DR: Introduces learnable graph patches as minimal semantic units for cross-domain graph pre-training.

AWARDS

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- **Second Place** in both base & large model subtracks of Red Teaming LLM@**NeurIPS 2023**, Torjan Detection Challenge(**Team leader**). [Code]
 - **National Scholarship (top 0.2% national-wide)** issued by Ministry of Education of the People’s Republic of China, 2021